Assignment 1

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Birds Eye View of Course

- Assignment
  - Simple Reactive Robot
  - Article + Video

- Project
  - Research Project
  - *“Contract”*
  - Presentation + Article + Video

See plan in Moodle (a plan is a plan…)

Rules

- Groups of ~<=4
- Often, the groups of the Assignment are the same as the group of the project
- Self Register in Moodle's groups
- Project "Public" Presentation (1st week of exams?)
- Grades are individual
- *Plan the **Assignment** for:*
  features (results) + article + demo_video
- *Plan the **Project** for : Intermediate present. +*
  features (results) + presentation +
  article + dissemination_video

Assignment
Simple Reactive Robot + Article

Deliverables:

1. ***Video + Demonstration(?) + Source code***
   of the developed agent
   (code, features, readme, etc.)

2. ***Article***
   (very minimal SoA, results, discussion, etc)
   [NOT a tech report]
   => Results

Assignment
Regarding Robot

(Start thinking about final project…)

- **Design and test a REACTIVE robot under ROS2**
  - 2D X,Y, Theta
  - Must have simple architecture (sensors, etc.)
  - Can NOT have ANY memory (except PID)
  - Can have a general, undetailed idea of the map
  - Can be subsumption architecture
  - Can have a PID controller

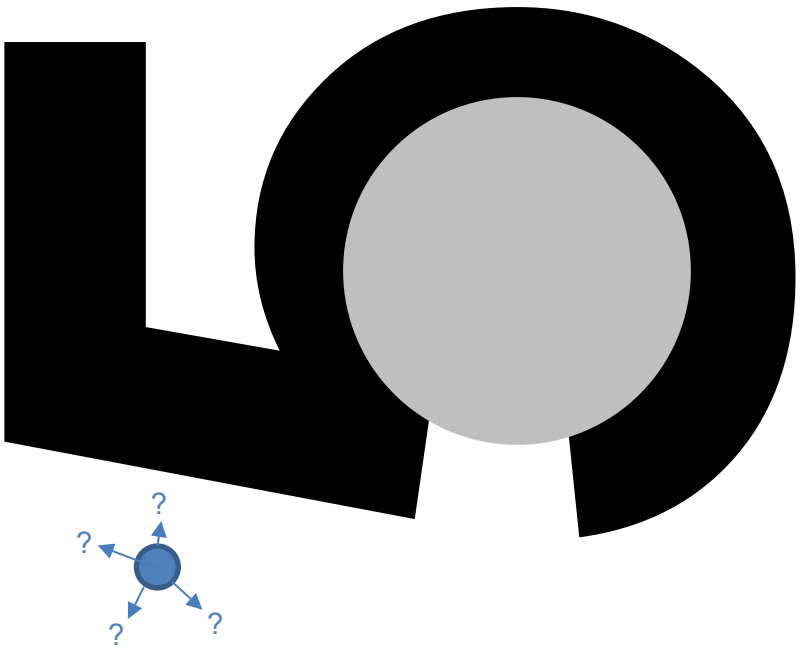
- No "cheating"
  - No privileged simulation information
  - No GPS nor global localization of any kind

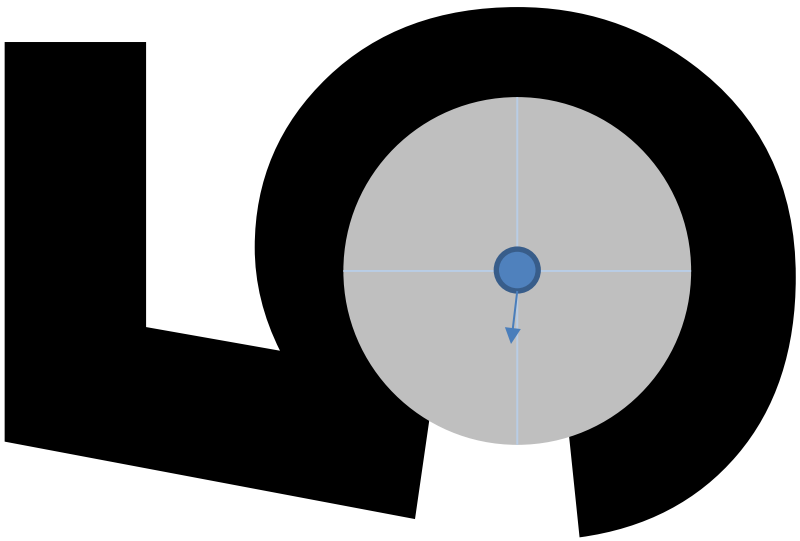
Assignment – Reactive Robot

Broad Goal:

- Robot starts outside of circle, random position and random heading
- Robot follows wall and when it detects circular walls, positions itself on the center of circle, heading out

- Rough Start Position (wall following mode) => End Position (center of circle)

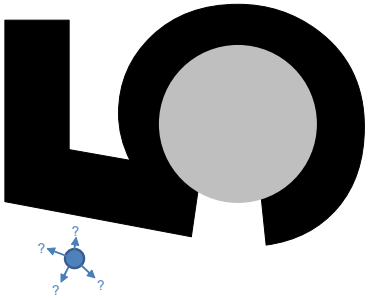


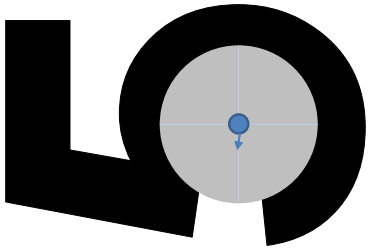


Assignment – Reactive Robot

- **Broad Goal:**
  - Robot starts outside of circle,
    random position and random heading
  - Robot follows wall and when it detects circular walls,
    positions itself on the center of circle, heading out
- **Robot:** Round, differential locomotion
- **Environment:** walls like a “5”, with inner circle
- **Sensors, actuators, speeds:** No restrictions!
- **Tech:** ROS + Simulator (Gazebo) + ...
- **Results for article**:

  Loop time, travelled path, performance,
  discuss stuck robot, control, performance improvement...

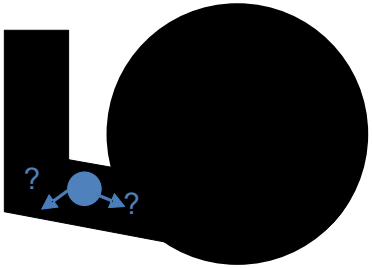


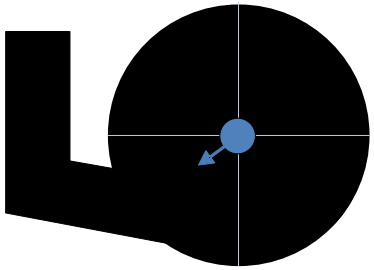


- In all situations, no memory at all (EXCEPT MAYBE PID)

Assignment – Reactive Robot - Extra Merit

- **Well written article - architecture + discuss limitations (map + robot + …)**
- Initial additional randomness, ex., if no walls wandering behaviour;
- Limited accel. and angular vel. (ω);
- Max. vel. & max wheel vel.;
- PI tuning;
- Table w/ mission times (const. ini. pose) and [PI(D), vel., accel, ang_vel, etc …]
- Additional tests when robot is inside "5"
  (gray becomes black and robot moves inside black) –
  – see figures on the right;
- More (simultaneous?) robots, larger robots,
  rectangular robots, trailer robots, etc;
- Simulated skidding, irregular walls outside of the circle;
- Trajectories (points over time);
- Different sensors (range, lidar with less rays, noise, sonars, etc.)
- Generalization of stop condition (AI? CNN?)
- Etc etc etc
- In all situations, no memory at all (EXCEPT MAYBE PI(D))





Assignment
Regarding Article

(Start thinking about final project…)

- **Write the article with:**
  - **Sample structure:** Abstract; Intro; (minimal) SoA; Robot's implementation and Architecture (control law); Experiments; Results and discussion; Conclusions and Future Work; Acknowledgments; References
  - **Template:** https://www.ieee.org/conferences/publishing/templates.html
  - (?) Target ICARSC conference
    - Similar to last years, IEEE template

Reference material: Regarding Robotics

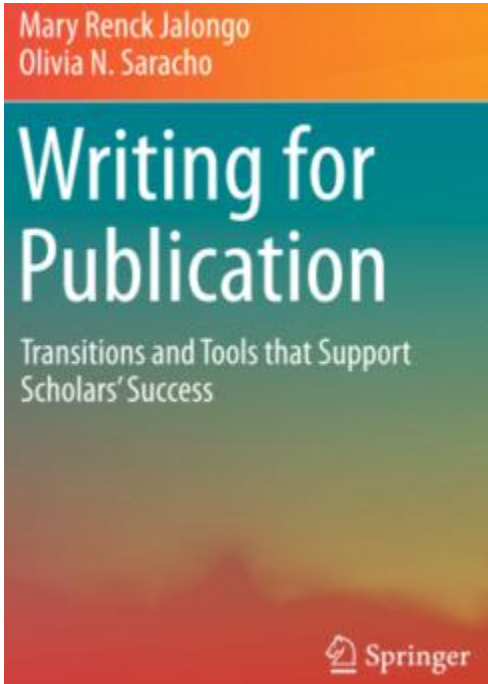
Recommended Readings:

- *"A Robust Layered Control System for a Mobile Robot", Rodney A. Brooks, IEEE*
- *Transactions on Robotics and Automation, 2(1), pages 14-23, April 1986.*
- *Part I – Robotic Paradigms of An Introduction to AI Robotics, Robin R. Murphy, Bradford*
- *Book, MIT Press, Cambridge, Massachussets, London England, 2000. ISBN:0-262-13383-0*
- *Behavior-Based Robotics, Ronald C. Arkin, MIT Press, 1998, ISBN 0-262-01165-4*

Reference material: Regarding the Article

Recommended Readings:

- https://xpertscientific.com/style-guide/
- https://www.springer.com/series/13812



Assignment

It is known that this work will have limitations!

- **Deliverable 1 - Code:**
  - ZIP the source code of the developed agent; include **readme.md** stating: directory structure; requirements (versions, etc); how to compile and how to execute
- **Deliverable 2 – Article:**
  - "Scientific" paper 3/4 pages - English language
  - Template - Standard A4 IEEE guidelines: http://www.ieee.org/conferences_events/conferences/publishing/templates.html
  - Submissions are NON-BLIND, that is, all author information should be explicit
  - Use formal references (as in template)
- **Deliverable 3 – Demo_Video** (basically the results)**:**
  - Video in single a separate file, under 100MB, 1080p, MP4
  - Sample workings, each case, Student's names, FEUP Logo

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